

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think> First, find how long it would take normally to download the file and then determine the impact of the restart.

- The file is 200 GB.
- Normally Carla can download at 2 GB / minute.
- The first 40% of the file is  $200\text{ GB} \times 0.40 = 80\text{ GB}$ .
- Time to download 80 GB at 2 GB / minute is  $80\text{ GB} / 2\text{ GB / minute} = 40\text{ minutes}$ .
- After the first 40%, a 20-minute restart occurs.
- The second 60% of the file is  $200\text{ GB} \times 0.60 = 120\text{ GB}$ .
- Time to download the next 120 GB at 2 GB / minute is  $120\text{ GB} / 2\text{ GB / minute} = 60\text{ minutes}$ .
- After this, the whole process restarts, meaning she repeats the process of 40 minutes down, then another 60 minutes, and then 20 minute restart.
- Calculate each completed cycle in the restart where each cycle is  $40 + 60 + 20 = 120\text{ minutes}$ .
- Combine the times in one cycle:  $40 + 60 + 20 = 120\text{ minutes}$ .
- The entire process is a series of these 120-minute cycles plus the 40 minutes at the beginning.

Final load time can be taken through repeated cycles:

- All the full 1.2 minute loops:

Each set of loops is 120 minutes.

- Within her waiting time x full cycles, the loop with rounding gives thus multiplied: one full loop out, 1 cycle and a continuous countdown as a summative recurring.

Thus summed, with the multiple-shifts large-leap:

- 5-cycle =  $5 \times 120\text{ mins} + 40\text{ minutes}$  ( final 0.875 to close to end ) [ 940 would be ]

Summarize and dispatch load: thus  
The exact configurable steps M:

- Thus, the total continuous sum, 120
- Summative.

Result:  
<answer> 680 </answer> minutes.  
Thus consumes repeated increment building giving cumulative. </think>